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SET-1

End Semester Examination 2024

Name of the Course: B.Tech. Biotechnology

Semester: 7th

Name of the Paper: Rational Drug Discovery

Paper Code: **TBT 702b**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question

Q1	(20marks)	
(a)	Define different steps of SBDD method for drug discovery.	CO1
(b)	Analyze the significance of Chou Fasman and Homology modeling methods in protein structure prediction.	
(c)	Highlight various methods used for drug target identification and validation.	
Q2	(20 marks)	
(a)	Differentiate between Quantum mechanics and molecular mechanics.	CO2
(b)	Correlate different parameters with equation of total force field? Justify the role of various force fields.	
(c)	Analyze the role of Energy minimization in Molecular structure prediction. Discuss any one type of Energy minimization algorithm.	
Q3	(20 marks)	
(a)	Discuss the algorithm of Molecular dynamics simulation? What do you mean by statistical ensembles?	CO3
(b)	Explain following file formats: 1) .trr (2) npt.tpr (3) .xvg (4) minim.mdp (5) .gmx	
(c)	Define steps of Principle component analysis? How do you apply it in MD simulations?	
Q4	(20 marks)	
(a)	Correlate Lipinski's rule of 5 with ADMET parameters of a lead molecule. Justify the role of Pubchem database.	CO4
(b)	Justify the significance of PyRx tool in virtual screening of molecules.	
(c)	Describe any one database used for searching lead molecules. What is significance of RMSD in molecular docking ?	
Q5	(20 marks)	
(a)	What are various molecular, geometrical and Hybrid descriptors used in QSAR modelling?	CO5
(b)	Describe Artificial Neural network and applications of ANN in drug designing.	
(c)	Define pharmacophore and application of Pharmacophore modelling in LBDD in detail.	

